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University of Information Technology & Sciences

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Lab Report

Course Title: Internet of Things Lab

Course Code: CSE 402

Submitted By:	Submitted To:
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Experiment No. 01:

Experiment Title: Introduction to Arduino UNO and blink a Light Emitting Diode (LED) with it.

Objectives:

Understand basic electronics: To explore the fundamentals of using an LED and an Arduino in a circuit for light control.

Learn Arduino programming: To gain hands-on experience in writing and uploading a program (sketch) to control an LED.

Examine circuit connections: To understand how to correctly wire an LED to an Arduino board and implement appropriate current-limiting resistors.

Control LED state: To investigate how digital pins on the Arduino can be used to turn an LED on and off in a periodic manner.

Test timing intervals: To experiment with different delay times between ON and OFF states of the LED and observe the effects on blinking frequency.

Validate program logic: To confirm the correct execution of a timing loop using `delay()` function in Arduino code to control the LED blinking pattern.

Introduce basic troubleshooting: To identify and resolve common issues in circuits and code that may prevent the LED from blinking correctly.

Introduction

The Arduino Uno is a popular open-source microcontroller board designed for learning and prototyping with electronics. It uses the ATmega328P chip and can be programmed through the Arduino IDE to interact with sensors, motors, and other electronic components.

Pin Diagram Overview:

- DigitalPins(0-13) : Used for input or output, with some supporting PWM (Pulse Width Modulation) for simulating analog signals.
 - AnalogPins(A0-A5): Read values from analog sensors, such as temperature or light sensors.
 - PowerPins :Includes 5V, 3.3V, GND(Ground) ,and Vin for power supply.
 - ResetPin : Resets the board.
 - SerialPins(TX,RX) : Used for communication between the board and other devices.
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Basic Knowledge:

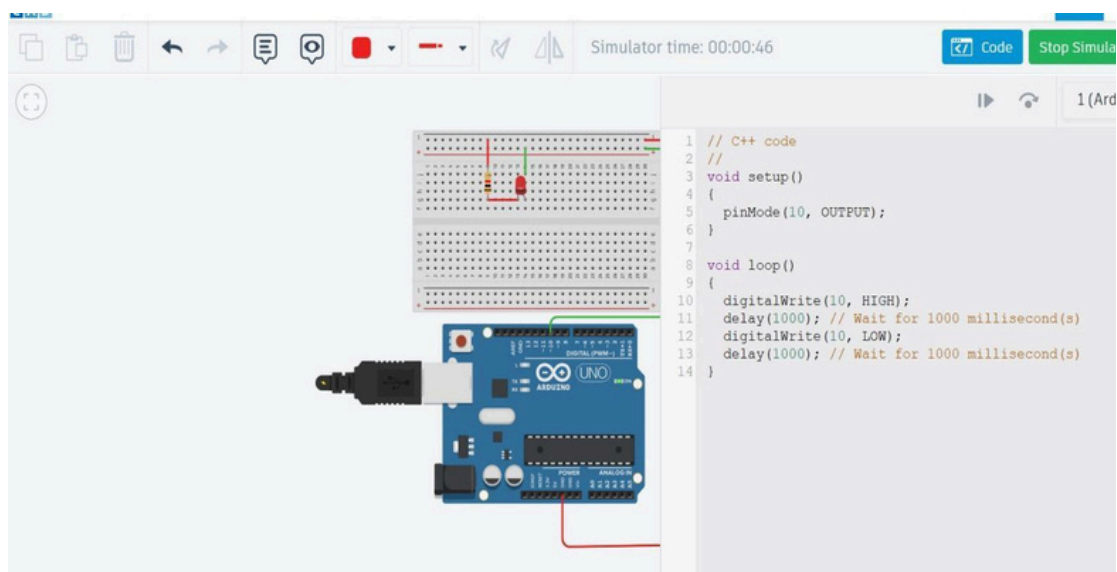
- Microcontroller: The ATmega328P chip processes instructions and controls connected components.
- ArduinoIDE: A platform for writing and uploading programs to the board, primarily using C/C++.
- PowerSupply: Can be powered via USB or an external source, with onboard regulation for stable voltage.

The Arduino Uno is an accessible tool for beginners to explore electronics and embedded programming.

Apparatus Required:

Name	Number
1. Breadboard-Small	1
2. Arduino Uno-R3	1
3. Red LED	1
4. Resistor-1K Ω	1
5. Jumper Wires	5
6. USB Power Cable	1

Circuit Diagram:



Working Procedure:

1. Create Circuit:

- Place Arduino UNO, LED, and 1kΩ resistor on Tinkercad workspace.
- Connect LED anode to pin 10 and cathode to resistor; resistor goes to GND.

2. Write and Upload Code :

- Open the Text Editor in Tinkercad, write the blinking LED code, and upload it to the virtual Arduino UNO.

3. Start Simulation :

- Start the simulation, and watch the LED blink every 1 second on pin 10.

4. Adjust Timing :

- Modify the delay in the delay(1000) line to change the blinking rate.

Code:

```
//      C++
code    //
void
setup() { }
void loop()
{ pinMode(10, OUTPUT);
  digitalWrite(10, HIGH);
  delay(1000); // Wait for 1000 millisecond(s)
  digitalWrite(10, LOW);
  delay(1000); // Wait for 1000 millisecond(s)
}
```

Discussion and Conclusion:

The loop() function runs continuously, turning the LED on and off with a set delay. The experiment demonstrates controlling the LED using digital signals (HIGH for ON, LOW for OFF).

The blinking LED experiment using the Arduino UNO is an excellent introduction to both electrical circuits and programming with microcontrollers. By understanding how to control an LED with an Arduino, beginners gain a fundamental understanding of how digital inputs and outputs work. The experiment also teaches the basics of timing control and how to use programming logic to control physical hardware. With the foundation of this simple project, users can begin exploring more complex applications, including sensor interaction, PWM control, and more.
